

Advanced techniques in applied economics

Lecture 7: Network interference and spillovers

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When treatment spills across a network

What changes today?

So far we treated each unit as if it lived in isolation. Now we ask what happens when one unit's treatment affects another unit's outcome.

Examples economists care about

- peer effects in schools,
- information spillovers in villages,
- diffusion of incentives across firms,
- contagion in financial and production networks.

Main difficulty

The treatment effect of unit i may depend not only on D_i , but also on neighbors' treatments.

Main message: graphs do not only describe dependence — they can also describe how treatment effects propagate.

Roadmap for today

1. potential outcomes and the role of SUTVA
2. what breaks when SUTVA fails
3. spillovers on graphs
4. exposure mappings and simple estimands
5. policy targeting under interference

Part 1: Potential outcomes and SUTVA

Quick review of potential outcomes

For each unit i and treatment level $d \in \{0, 1\}$, define

$$Y_i(d)$$

as the outcome that would be observed under treatment status d .

Average treatment effect

$$\tau = \mathbb{E}\{Y_i(1) - Y_i(0)\}.$$

The classical picture

Each unit has exactly two relevant potential outcomes: one under treatment and one under control.

Interpretation

This works well when units do not interfere with each other.

SUTVA

SUTVA stands for the **Stable Unit Treatment Value Assumption**.

Two parts

- **No multiple versions of treatment:** treatment $D_i = 1$ means the same thing for all units.
- **No interference:** unit i 's outcome depends only on its own treatment, not on others' treatments.

Under SUTVA

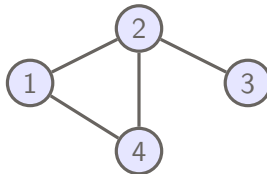
$$Y_i(\mathbf{d}) = Y_i(d_i),$$

so the potential outcome depends only on the unit's own treatment.

SUTVA is what lets us write $Y_i(1)$ and $Y_i(0)$ without worrying about the rest of the population.

Why SUTVA can fail

Suppose students are connected by a friendship network.



Example

If unit 2 receives tutoring, then unit 1 may benefit indirectly through peer interaction.

Then

$$Y_1(\mathbf{d})$$

depends on the whole assignment vector $\mathbf{d}$, not only on d_1 .

What changes when SUTVA fails?

Without SUTVA, the potential outcome is indexed by the whole treatment assignment:

$$Y_i(\mathbf{d}), \quad \mathbf{d} = (d_1, \dots, d_n).$$

Combinatorial explosion

With n units and binary treatment, there are

$$2^n$$

possible assignments, so each unit may have an enormous number of potential outcomes.

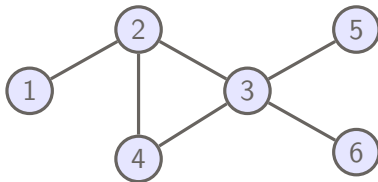
Main challenge

We need structure to reduce this complexity. Networks and exposure mappings are one way to do it.

Part 2: Spillovers on graphs

A graph-based view of interference

Let G be a graph describing who can affect whom.



Interpretation

The graph restricts the set of possible spillovers: unit i may be affected by treatments assigned to its neighbors, but perhaps not by distant units.

This is the key modeling move

The network is what makes interference manageable.

Exposure mappings

A common simplification is to replace the full assignment vector by an exposure summary.

Example

Instead of writing $Y_i(\mathbf{d})$, write

$$Y_i(d_i, e_i),$$

where e_i is an exposure measure such as:

- number of treated neighbors,
- share of treated neighbors,
- whether at least one neighbor is treated.

Interpretation

Two treatment assignments that produce the same (d_i, e_i) are assumed to generate the same potential outcome for unit i .

Direct and spillover effects

If exposure is summarized by e_i , we can define several effects.

Direct effect

Compare

$$Y_i(1, e) - Y_i(0, e),$$

holding neighbor exposure fixed.

Spillover effect

Compare

$$Y_i(d, e') - Y_i(d, e),$$

holding own treatment fixed.

Under interference, “the treatment effect” splits into a direct component and a spillover component.

A village example

Suppose treatment is access to information about fertilizer.

- $D_i = 1$ means household i receives the information directly.
- Neighbors who receive the information may also affect household i through discussion or imitation.

Potential outcomes

A natural model is

$$Y_i(d_i, e_i),$$

where e_i is the proportion of treated neighbors.

Interpretation

- direct effect: effect of being treated yourself,
- spillover effect: effect of living in a more treated neighborhood.

A school or peer-effects example

Suppose treatment is assignment to a tutoring program.

Own-treatment effect

Does the program improve the treated student's outcome?

Peer spillover

Does having more treated friends improve the untreated student's outcome?

Why this matters

A policy may produce benefits even for untreated units. If we ignore that, we may badly mis-measure policy impact.

Part 3: *A* tiny simulation

A toy spillover simulation in R

Suppose outcomes satisfy

$$Y_i = 1.0 D_i + 0.8 E_i + \varepsilon_i,$$

where E_i is the share of treated neighbors.

```
set.seed(1)
n <- 50
A <- matrix(0, n, n)
for (i in 1:(n-1)) {
  A[i, i+1] <- 1
  A[i+1, i] <- 1
}

D <- rbinom(n, 1, 0.4)
deg <- rowSums(A)
E <- (A %*% D) / pmax(deg, 1)
Y <- 1.0 * D + 0.8 * E + rnorm(n)
```

```
summary(lm(Y ~ D))
summary(lm(Y ~ D + E))
```

Interpretation

If we regress only on D , the spillover term is omitted. Including an exposure measure E lets us separate the direct and spillover components.

What this simulation shows

The graph does not only generate dependence. It also determines how treatment reaches other units.

Lesson

If spillovers are real, then an analysis that ignores exposure may:

- overstate or understate the direct effect,
- miss important social multipliers,
- and suggest the wrong policy ranking.

Interference turns a treatment assignment problem into a network assignment problem.

Part 4: Targeting and policy evaluation under interference

Why targeting is harder under interference

Without interference, a planner can ask:

Who benefits most from treatment?

With interference, the planner must instead ask:

Which pattern of assignments creates the most social value on the network?

Main difference

Treating one unit may help or hurt others. So the best policy depends on the whole assignment, not only on unit-by-unit treatment effects.

Davide Viviano as a modern anchor

A nice modern entry point is Davide Viviano's work on policy targeting under network interference.

Main idea

Instead of treating units independently, choose treatment assignments to maximize average welfare when spillovers are present.

Why this is attractive for applied economics

- treatment effects may be heterogeneous,
- networks may matter for welfare,
- policy design becomes an optimization problem under interference.

A very simple targeting picture

Suppose a treatment has both direct and spillover benefits. Then a planner may prefer to treat

- central nodes,
- bridges between groups,
- or units whose treatment benefits many untreated neighbors.

Contrast with the no-interference case

Without spillovers, targeting is mostly about the treated unit. With spillovers, targeting is also about the unit's position in the network.

Packages and practical tools

There is no single dominant package for all interference problems, but there are useful tools.

R

- RCT2: two-stage randomized designs with interference,
- latenetwork: IV-style estimation under network interference,
- igraph: graph construction and network summaries.

Python

For general graph handling, `networkx` is useful. For causal spillover estimation, practice is more fragmented and often custom.

Practical message

For teaching and small projects, it is often best to combine a simple exposure mapping with standard regression or randomization-based logic.

A package-style example in R

For two-stage randomized designs with interference, RCT2 and latenetwork provide useful starting points.

```
# RCT2: two-stage randomized designs  
library(RCT2)
```

```
# see package vignette for the india  
# insurance example  
# rand <- CADERand(india, 0.95)
```

```
# latenetwork: unknown interference  
# with IV-style estimands  
library(latenetwork)
```

```
# data <- datageneration(n = 2000)  
# result <- direct(Y=data$Y, D=data$D,  
# Z=data$Z, A=data$A,  
# S=rep(TRUE, 2000))
```

Interpretation

These packages are useful anchors, but the main conceptual work is still in specifying the network and the exposure mapping.

Take-away

- Under SUTVA, each unit's potential outcome depends only on its own treatment.
- Under network interference, outcomes may depend on the full assignment vector.
- Exposure mappings use graph structure to simplify the problem.
- Direct and spillover effects are conceptually distinct.
- Policy targeting under interference is about assignments on a graph, not isolated units.

Interference is one of the clearest examples of how graphs enter applied economics not only descriptively, but directly in identification and policy design.

Looking ahead

Next lecture

We now shift from *who affects whom* on a network to *how the sign of dependence itself can be exploited*.

Bridge forward

So far graphs helped us organize dependence and causal structure. Next we will ask what happens when dependence is known to be positive.